

Course Syllabus
Assumption University
Vincent Mary School of Engineering, Science and Technology
Bachelor of Science Program in Computer Science (BSCS)
Semester 2/2025

COURSE INFORMATION	
Course Number:	CSX3008
Course Title:	Operating Systems
Credits:	3
Pre-requisite:	None
Status:	Major Required Course
Classroom:	VMES
Day and Time:	Monday (09:00-12:00), and Tuesday (13:30 -16:30)
Material Portal:	MS Team File

COURSE INSTRUCTOR	
Instructor:	Asst. Prof. Dr. Anilkumar K.G
Email & Mobile No:	Office: VMES0401, aGopalakrishnan@au.edu , 0891351711

COURSE OUTLINE																																							
Course Description:	Components and functions of operating systems, including uni-programming, multi-programming, multitasking, and multithreading, as well as resource management functions, process scheduling algorithms, device management algorithms, virtual memory management, and classical problems related to operating systems, such as deadlock, starvation, and concurrency.																																						
Learning Objectives:	Upon completion of this subject, students should be able to: 1. To know the components and functions of an operating system 2. To have the ability to analyze the issues related to operating systems 3. To work out the tradeoffs involved in designing a modern operating system																																						
Learning Contents:	<table> <tr> <th>Week</th><th>Topic</th></tr> <tr><td>01</td><td>Operating System Overview</td></tr> <tr><td>02</td><td>Process Description and Control</td></tr> <tr><td>03</td><td>Process Description and Control(continue)</td></tr> <tr><td>04</td><td>Threads management</td></tr> <tr><td>05</td><td>Threads management (continue)</td></tr> <tr><td>06</td><td>Process Synchronization</td></tr> <tr><td>07</td><td>CPU scheduling</td></tr> <tr><td></td><td>Programming Assignment 1 Presentation</td></tr> <tr><td>08</td><td>Quiz I</td></tr> <tr><td>09</td><td>Midterm Examination</td></tr> <tr><td>10</td><td>Deadlock</td></tr> <tr><td>11</td><td>Memory Management</td></tr> <tr><td>12</td><td>Virtual Memory</td></tr> <tr><td>13</td><td>Input and Output management</td></tr> <tr><td>14</td><td>Programming Assignment 2 Presentation</td></tr> <tr><td>15</td><td>Programming Assignment 3 Presentation</td></tr> <tr><td></td><td>Quiz II</td></tr> <tr><td></td><td>Final Examination</td></tr> </table>	Week	Topic	01	Operating System Overview	02	Process Description and Control	03	Process Description and Control(continue)	04	Threads management	05	Threads management (continue)	06	Process Synchronization	07	CPU scheduling		Programming Assignment 1 Presentation	08	Quiz I	09	Midterm Examination	10	Deadlock	11	Memory Management	12	Virtual Memory	13	Input and Output management	14	Programming Assignment 2 Presentation	15	Programming Assignment 3 Presentation		Quiz II		Final Examination
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COURSE MATERIALS	
Main Textbook:	Operating System Concepts , 10th Edition, Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Wiley (ISBN: 978-1-118-09375-7)
Supplementary Textbook:	Operating Systems: Internals and Design Principles , William Stallings, Prentice Hall (ISBN: 978-0-133-80591-8)

COURSE POLICIES	
1.	Students must have at least 80% of class attendance to be eligible for the final examination.
2.	If a student is absent from more than three classes <u>without any relevant reason</u> , then they should withdraw from the course immediately.
3.	The student who misses the final exam will receive a grade based on their total marks. In this case, the 'F' grade (if any) will be replaced with a 'W.'

ASSESSMENT APPEAL POLICY	
For any assignments/projects and examination(s) (EXCLUDING final examination), the lecturer will announce scores and discuss with students about solutions approximately within 1-3 weeks after the submission deadline and after finishing grading. Students may request the lecturer for an assessment appeal, if any, within 1 week or as specified by the appeal's deadline.	

PROGRAMMING ASSIGNMENT OUTLINE			
Programming Assignments:	No.	Instruction	Marks
	1	Write a simulation program to simulate the behavior of FCFS and Round Robin CPU scheduling algorithms (the simulations must be based on the <i>arrival time</i> of each process).	8%
	2	Write a simulation program to show the behavior of a banker's algorithm for multiple resource unit allocation problems. The program should indicate whether a solution ended in a safe state or not. Also, the program should be able to collect input datasets from either a text file or a random generator.	8%
	3	Simulate the following page replacement algorithms: OPTIMAL, LRU, and FIFO.	8%
	Total		24%
Programming Assignments Evaluation Criteria:	Grading of a presented assignment is based on the following criteria: The program accepts all inputs. Ultimately, the Program shows the progress of the algorithm step-by-step The program offers all outputs correctly The program provides a clean and simple UI design (A hard copy of the full source code of each programming assignment must be submitted)		